

Input Form (IF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MathDial is English text-only [Q1, Q32], and the deployment context is confirmed English text-only on mobile/enterprise platforms with no code-switching, script variation, or vernacular content. There is no modality, script, or language-encoding mismatch. Dataset analysis confirms all 18 sampled examples are pure English prose with no Hindi, Hinglish, or transliteration [DATASET-D6, DATASET-D15]. A safety filter using Perspective API removes toxic content [Q157, Q158]. Minor surface-level imperfections in teacher utterances (spelling/typing errors) [DATASET-D19] are noted but do not affect form validity. The deployment is Indian metro with English-medium instruction, and metro infrastructure supports the input modality [WEB-3, WEB-4, WEB-8].

ID	Evidence
Q1	"We describe how we use this framework to collect MATHDIAL, a dataset of 3k one-to-one teacher-student tutoring dialogues grounded in multi-step math reasoning problems." (p.1)
Q32	"We use InstructGPT (text-davinci-003) (Ouyang et al., 2022) to generate student turns." (p.5)
Q157	"As we are interested in real educational use cases for our tutoring system, we apply a safety filter to filter out conversations with any sensitive content." (p.18)
Q158	"In particular, we use the Perspective API to filter out conversations containing toxic content (<1%)." (p.18)
WEB-3	Delhi has 3,000+ English-medium schools (https://ezyschooling.com/english-medium-schools-in-delhi)
WEB-4	Delhi government Schools of Excellence operate in English medium (https://www.schoolmykids.com/schools/list-of-schools-of-excellence-in-delhi-india)
WEB-8	National school internet connectivity 63.5% per UDISE+ 2024-25 (https://educationforallinindia.com/analysis-of-udise-2024-25-data-by-prof-arun-c-mehta/)
D6	Jia: Sure. I started by figuring out Maxwell's age in 2 years... So, if Maxwell is twice her age in 2 years, he will be 4 years old. Then, I subtracted 2 from 4 to get the answer that Maxwell is currently 2 years old. — Simulated student gives detailed, multi-sentence explanation of reasoning — verbose, articulate, reasoning-out-loud style
D15	question: Cecil has \$600. Catherine has \$250 less than twice as much as Cecil... — US-dollar amounts and US-style name inventory (Cecil, Catherine, Carmela); no Indian cultural grounding
D19	Teacher: Luca do you think youu could check your answer again please? maye try and think..." / "Teacher: was the total three trimes the amount oof his contribution before oor after he contributed? — Multiple spelling and typographic errors in teacher utterances; affects naturalness of benchmark dialogue

Strengths

- Perfect form alignment: both benchmark and deployment are English text-only [Q1, Q32]
- All sampled dialogues confirmed monolingual English with no script issues [DATASET-D6, DATASET-D15]
- Safety filtering applied to remove toxic content [Q157, Q158]
- Metro infrastructure supports text-based interfaces [WEB-3, WEB-8]

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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: No signal-distribution mismatch. Both source and target use English Latin-script natural language text [Q1, Q32, DATASET-D15].

IF-2: Yes — Indian metro infrastructure supports text-based AI tutoring interfaces; Delhi has 3,000+ English-medium schools [WEB-3] and Government Schools of Excellence operate in English medium [WEB-4]. National school internet connectivity is 63.5% [WEB-8] with metros expected higher.

IF-3: No domain-specific form differences. The benchmark and deployment both operate in text-based open-ended dialogue.

IF-4: No form mismatches identified. Minor surface noise in teacher utterances (typos) [DATASET-D19] is naturalistic and does not harm external validity for the deployment.

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